

Here is “intermediate feedback” on your group project work to date. All the feedback applies to most groups. You should store this feedback in some shared Group Project folder, and access it later so that your future project modules reflect the feedback.

Landscapes:

- **Misunderstanding the nature of landscapes:** Many groups fundamentally misunderstood certain aspects of the performance landscape idea as about sequential plans. It is about spatial interdependence. Navigating to higher elevations in the landscape does not mean taking a sequence of actions in the sense of the maze metaphor. Sequence concerns the adaptive process of trying out different sets of spatially-interdependent modules or variables. The “high fitness” solution should be a blueprint-style plan that, *in turn*, supports a user of your system to take many different sequences of actions depending on their situation.
 - o Also: variations do not equal variations based on different measures of quality. X-axes in the landscape are actions, NOT different performance criteria (what you have now). We assume (for now) that those performance criteria are aggregated under a single y-axis performance variable.
- **Narrowly understanding the topology of landscapes:** Most groups depicted a rugged landscape. This is understandable: it is easy to think of some local peaks and a global peak. Indeed, AI and organizational researchers have often thought this way too. But I would challenge you: recall that a rugged landscape already assumes that you have mostly solved or understand the problem to be solved. Here’s some other points of view to consider:
 - o Single-peaked: note that this is the dominant view of the landscape in the AI field today (except it is almost always thought of in terms of “gradient descent” — of minimizing some prediction error). You could also look at your system as a set of modules that can be easily separated, searched, or predicted, and therefore able to be optimized through gradient descent (more about this next week in class).
 - o Mesa: this is the original view of the fundamental landscape in the AI field: it may be more realistic for thinking about how a student goes from no familiarity with new AI ideas to some satisfactory understanding of the concepts.
- **Depicting landscapes as artwork:** Example (done by many groups): “We asked Gemini to enhance our whiteboard sketch for a more pleasing visual representation”. Yet the visual examples of many groups provide meaningless detail and artistic flourishes. Instead, what matters for your group (for now, at least) is the basic topology. Let’s make a rule for our projects: only represent landscapes in 2D, and no artwork! What you will eventually need to define is how is fitness measured, *and* what does it mean to move up or down 1 unit of elevation in the landscape. This all needs to be describe precisely in terms of symbol manipulations. Artwork doesn’t matter.

LLM use:

- LLMs are really powerful. Yet ultimately, they are a small part of AI ideas. Using them the wrong way or excessively will not be a good strategy for your project. *Some heuristics to guide your own group's work on your use of LLMs:*
- **Could someone generate basically the same thing by casually prompting an LLM?** If so, then your system will not really add value in a business setting. Your value will come instead from thinking deeply at the symbolic-level about how to architect the whole thing. Hence, consider if what you are proposing reflects deep and precise understanding of the problem. Note: this understanding can be tentative and wrong (that is natural) as you try to build your system, but it should always reflect deep thinking by your group, resulting in insights (right or wrong) that are not obvious.
- **Could your system be implemented in a valuable way without an LLM? (e.g., the “when2meet.com test”)**. If so, this is a good sign you are onto something. Consider: all of the groups came up with different modules, operators, heuristics, etc. How to select among them, and figure out how to put them together into a coherent system? If you ask an LLM to do this for you, it can give you thousands of plausible ways. Your group's added value is to be the “architect” that selects just the right building blocks and how to put them together.

Modules and evolution:

- **Generating variation in modules:** Many groups misunderstood how variation is generated in evolutionary strategies. It is not primarily variation in sequences, but rather in *module architectures*; sequential aspects come from the heuristics, so this is part of it, but not the whole thing. So variation should be in what the modules are, and how they are specified, not in how they are ordered like some pin factory tasks. Studying AI ideas (or any complex topic) is an iterative process, not just or even primarily a sequential one. Likewise: user feedback should be on variations in modules in terms of their type and content. Not sequence, not just flowcharts.
 - o The key thing about the modules for now is to start to describe the problem more abstractly in spatial terms. What many groups have described is variation on sequential plans — “if A, use module B; If C, use module D”. Instead, think of modules spatially: all or part of modules can be invoked for a particular state of a problem. Think of it in terms of the watch example: watch components are not invoked one at a time, but all of them are needed for the watch to function.

States:

- **States are descriptions, not effects:** Most groups did not clearly describe states. States are descriptions of a problem, from the problem-solver's point of view, at a particular point in solving the problem. Some groups instead focused on *effects* of being in a state: e.g., aspiration level.

- An example: a claim that the “state” of a conceptual network could be when: “the determined aspiration level of understanding is reached”. This is an outcome of a state, not the state itself. You need a description of what the problem looks like when an aspiration level is reached.
- **States eventually need to be described at symbol-level precision:** Simply put: a state should be *what the symbols in your system look like to a student at a particular point in their process of using your system*. Very few of the operators proposed by any group indicate *what happens to the state* — this is in some sense natural, since we have not covered the “Representation” topic yet, but from now on, you should keep this at the forefront of your focus. What does a student actually see? How is this updated? What happens to any memories?
 - Hence, need much more precision about the **symbols** — this course: finding a middle ground — NOT Python code, but not just casual descriptions either... most of these were written casually from almost a user behavior point of view (“I want a quiz”), or too much of a coder’s view; instead, need to think in terms of **precise symbols**.
- **Operators and their effects on state:** Many groups focused on extracting concepts or similar things, and perhaps ranking them by importance. This is a quite standard approach in currently popular AI. But it is ultimately not close to good enough. What is also needed is an understanding of relations (interdependencies) among the concepts, and at various levels of abstraction. You will in subsequent modules need to be far more precise: what is meant by “skim”? “Summarize”?
 - These need to be operationalized in symbol-manipulation terms. To do so, dig down into your information processing assumptions: are the symbolic units bit strings, lists, something else (words, concepts, phrases, etc.)? What is a “concise” summary? A sentence? One page? Why?
- **States accumulate – need to account for this:** Key point about state: like short-term memory: should be *relative to the description needed by a problem-solver*. Hence, it depends on the current understanding of the current problem-solver. Some students will get ideas right away, others need to see them from many points of view. Key here is abstraction (hiding information): knowing when and how to do this.
 - Recall Minsky: a big idea was that you would have a society of agents (perhaps corresponding to fine-grained modules), and they would maintain a society of *memories* — these might be like fine-grained descriptions of the states of a problem. Yet, most groups’ proposals were more about one-off outputs (prompt → summary etc.). There was little precision about what the outputs should look like, nor how they might be stored as memories and combined and manipulated later on.

Fitness criteria:

- **Going beyond genetic approaches:** Most groups used a genetic approach, even if they did not realize it — the way they approached fitness is: “let us find some way to

randomize something based on some benchmark and therefore generate different variations, then test people's reactions". This is natural since we are still early in the project; yet a better way is to build a deep functional understanding of your modules, so that your system can evolve not just through random trials (e.g., A/B testing), but through objective constraints on modules.

- To see the big picture here, consider this course's quizzes as a very rough and not so ideal way of assigning credit to your own performance. Each week, we have a quiz where you try to optimize your score. But what employers really want are students who, over months and years, have accumulated a deep understanding of AI ideas and how to build a complex AI system. So consider deeply the point we made last week: *fitness* (i.e., a high quiz score) *does not equal intelligence* — it is at best a useful but rough proxy for it. And even then, fitness will usually be best thought of as a way of assigning credit regarding a specific module or goal.
- Constraints: you will need to turn operators/modules etc. into local goal states — which will need *specific criteria* (we will call these **constraints**). This will be the top for our next session (and many future sessions).